F13-SW-103
2000-F13-SLD-EL-10911

F13-SR-001
415V SAG MILL SWITCHROOM

SWR EARTH BAR

FEEDER FROM F13-SW-002
2000-F13-SLD-EL-10301

TO F13-SW-002
2000-F13-SLD-EL-10301

F13-SW-102
SAG MILL 415V ESSENTIAL SWITCHBOARD

EARTH BAR

NEUTRAL BAR
NOTE 1

AFD

2A
63A

2A
63A

MANUAL
MCCB CHANGEOVER
SWITCH 3 POLE 630A

00F-05

415V 3NE~ 50Hz 630A 65kA for 1sec

INC

01F-16 160A 2A 32A / 90A / 160A 250A
01F-05 32A 2A 32A / 70A / 32A 32A
01F-21 250A 2A 32A / 90A
01F-12 100A 2A 32A / 70A
01F-09 63A 2A 32A / 70A
02F-16 250A 2A 32A / 90A
02F-11 250A 2A 32A / 90A / 160A 250A
02F-06 250A 2A 32A / 90A / 160A 250A

ECR

69kW	10kW	10kW
F13-DB-001 SAG MILL 415V SWITCHROOM L&SP DB 2000-F13-SCM-EL-10305	F13UPS001-ISL 240VAC UPS MAINTENANCE BYPASS SWITCH 2000-F13-SCM-EL-10306	F13-DB-201 SAG MILL 415V SWITCHROOM 240VAC UPS DB 2000-F13-TLD-EL-10012	F13-UPS-001 SAG MILL 415V SWITCHROOM 240VAC UPS 2000-F13-TLD-EL-10012

69kW	48kW	15.9kW	69kW	69kW	69kW	
F13-DB-304 MILLING AREA 1 L&SP DB 2000-F13-SCM-EL-10307	F13-CR-402 CLASSIFICATION AREA OVERHEAD CRANE 2000-F13-SCM-EL-10308	F13-CR-411 MILL LINER MAINTENANCE HOIST 1 2000-F13-SCM-EL-10309	F13-DB-307 MILLING AREA 4 L&SP DB 2000-F13-SCM-EL-10364	F13-DB-308 MILLING AREA 5 L&SP DB 2000-F13-SCM-EL-10973	F13-DB-003 SAG & BALL MILL VSD SWITCHROOM L&SP DB 2000-F13-SCM-EL-10906	

NOTES

1. EARTH-NEUTRAL LINK INSTALLED IN UPSTREAM F13-SW-002 INCOMER TIER. SIGNAGE SHALL BE PLACED ON F13-SW-102 SWITCHBOARD READING "M.E.N. LINK LOCATED IN F13-SW-002 INCOMER TIER"
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
4. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.

LEGEND

PFR	PHASE FAILURE RELAY
I >	EARTH LEAKAGE RELAY
xxxA / xxxA	ST TRIP SET POINT / CB SIZE
AFD	ARC FLASH DETECTION
ECR	EARTH CONTINUITY RELAY

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 15/09/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

2000-F13-SLD-EL-10311_Rev1

F13-SW-102 415V essential switchboard - single line diagram

D:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\E\LF13 - MILLING & CLASSIFICATION\2000-F13-SLD-EL-10311.DWG 27AUG25 7:52AM HOTSEAT04